The CSA Operator Is Unitary

Setting. Echo $\mathbf{Y}$ and image $\mathbf{H}$ are complex $M \times N$ matrices (M azimuth, N range samples). vec$(\cdot)$ stacks the columns into one vector,

$$\mathbf{y} = \mathrm{vec}(\mathbf{Y}) = \begin{bmatrix} \mathbf{Y}_{:,1} \\ \vdots \\ \mathbf{Y}_{:,N} \end{bmatrix} \in \mathbb{C}^{MN}, \qquad \mathbf{h} = \mathrm{vec}(\mathbf{H}) \in \mathbb{C}^{MN}, \tag{1}$$

which only rearranges entries, so $\|\mathbf{Y}\|_F = \|\mathbf{y}\|_2$. CSA [1] is linear, so it is a matrix $\mathbf{\Phi}$: $\mathbf{h}_{\mathrm{CSA}} = \mathbf{\Phi}\mathbf{y}$.

CSA in matrix form. Azimuth FFT, chirp scaling, range FFT, range compression and bulk RCMC, range IFFT, azimuth compression and residual phase compensation, azimuth IFFT:

$$\mathbf{\Phi} = \mathbf{F}_a^{-1}\mathbf{P}_3\,\mathbf{F}_r^{-1}\mathbf{P}_2\,\mathbf{F}_r\mathbf{P}_1\,\mathbf{F}_a, \tag{2}$$

- $\mathbf{F}_a = \mathbf{I}_N \otimes \mathbf{F}_M$: azimuth FFT (within each column); $\mathbf{F}_r = \mathbf{F}_N \otimes \mathbf{I}_M$: range FFT (within each row); $\mathbf{F}_K$: K-point DFT matrix, $\otimes$: Kronecker product.
- $\mathbf{P}_1$: chirp scaling; $\mathbf{P}_2$: range compression and bulk RCMC; $\mathbf{P}_3$: azimuth compression and residual phase compensation.

A product of unitary matrices is unitary, so it suffices that every factor in (2) is.

Phase multiplications $\mathbf{P}_1, \mathbf{P}_2, \mathbf{P}_3$. Each is a Hadamard product with a unit-magnitude array, e.g. $\mathbf{X}_{\mathrm{out}} = \mathbf{E}_1 \odot \mathbf{X}_{\mathrm{in}}$ with $[\mathbf{E}_1]_{m,n} = e^{j\phi_1(f_{a,m},\tau_n)}$, ϕ_1 real. After vectorization,

$$\mathbf{P}_1 = \mathrm{diag}\big(\mathrm{vec}(\mathbf{E}_1)\big) = \mathrm{diag}\big(e^{j\phi_{1,1}}, \ldots, e^{j\phi_{1,MN}}\big) \;\Rightarrow\; \mathbf{P}_1^H\mathbf{P}_1 = \mathrm{diag}\big(|e^{j\phi_{1,1}}|^2, \ldots, |e^{j\phi_{1,MN}}|^2\big) = \mathbf{I}; \tag{3}$$

$\mathbf{P}_2, \mathbf{P}_3$ differ only in the phase function.

DFTs $\mathbf{F}_a, \mathbf{F}_r$. The normalized $\mathbf{F}_K$ is unitary, and a Kronecker product of unitary matrices is unitary. An unnormalized forward FFT (factor $\sqrt{K}$) with an inverse FFT divided by K (factor $1/\sqrt{K}$) is also fine: each axis has one FFT and one IFFT, so the factors cancel.

Conclusion.

$$\mathbf{\Phi}^H\mathbf{\Phi} = \mathbf{\Phi}\mathbf{\Phi}^H = \mathbf{I}, \qquad \mathbf{\Phi}^{-1} = \mathbf{\Phi}^H. \tag{4}$$

Consequence: the echo-domain fidelity equals the image-domain fidelity. With the echo model $\mathcal{A} = \mathcal{I}^{-1}$ (inverse CSA), $\mathrm{vec}(\mathcal{A}(\mathbf{H})) = \mathbf{\Phi}^{-1}\mathbf{h}$ and $\mathbf{y} = \mathbf{\Phi}^{-1}\mathbf{h}_{\mathrm{CSA}}$:

$$\begin{aligned}
\big\|\mathbf{Y} - \mathcal{A}(\mathbf{H})\big\|_F^2 &= \big\|\mathbf{y} - \mathbf{\Phi}^{-1}\mathbf{h}\big\|_2^2 && \text{(vectorization preserves the norm)} \\
&= \big\|\mathbf{\Phi}^H(\mathbf{h}_{\mathrm{CSA}} - \mathbf{h})\big\|_2^2 && (\mathbf{y} = \mathbf{\Phi}^{-1}\mathbf{h}_{\mathrm{CSA}},\ \mathbf{\Phi}^{-1} = \mathbf{\Phi}^H) \\
&= (\mathbf{h}_{\mathrm{CSA}} - \mathbf{h})^H\mathbf{\Phi}\mathbf{\Phi}^H(\mathbf{h}_{\mathrm{CSA}} - \mathbf{h}) && (\|\mathbf{v}\|_2^2 = \mathbf{v}^H\mathbf{v}) \\
&= \big\|\mathbf{h}_{\mathrm{CSA}} - \mathbf{h}\big\|_2^2 = \big\|\mathbf{H}_{\mathrm{CSA}} - \mathbf{H}\big\|_F^2 && \text{(by (4))}.
\end{aligned} \tag{5}$$

So a sparse reconstruction from the echo is the same problem as one on the CSA image, which separates into per-pixel problems. Invertibility alone is not enough: for a non-unitary $\mathbf{\Phi}$ the fidelity is the weighted norm $\|\mathbf{\Phi}^{-1}(\mathbf{h}_{\mathrm{CSA}} - \mathbf{h})\|_2^2$, which couples the pixels.

References

[1] I. G. Cumming and F. H. Wong, *Digital Processing of Synthetic Aperture Radar Data: Algorithms and Implementation*. Norwood, MA, USA: Artech House, 2005.